

COMPUTER SECURITY (320)

NATIONAL – 2019

Knowledge Test:

Certification Exam: MTA Security Fundamentals _____ (100 points)

Objective Test:

Objective Test: _____ (500 points)

TOTAL POINTS _____ ***(600 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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Workplace Skills Assessment Program competition.

Identify the letter of the choice that best completes the statement or answers the question. Mark “A” if the statement is true. Mark “B” if the statement is false.

- 1) Error number _____ is used to signal a server error in HTTP.
 - a) 200
 - b) 500
 - c) 404
 - d) 300

- 2) The “C” in the information security CIA triad stands for _____.
 - a) Callability
 - b) Customizability
 - c) Conformability
 - d) Confidentiality

- 3) Which of the following copies traffic from all ports to a single port, and disallows bidirectional traffic on that port
 - a) Straddling
 - b) Port redistribution
 - c) Port spanning
 - d) Socket mixing

- 4) Which type of attack uses a constant flood of packets to prevent normal use of a network resource?
 - a) Fork bomb
 - b) Branching bomb
 - c) Spider attack
 - d) DoS

- 5) Which of the following is an example of a directory access protocol?
 - a) IDS
 - b) FTP
 - c) LDAP
 - d) OSI

- 6) This protocol operates on 2.4 GHz and has a bandwidth of 1Mbps or 2Mbps.
 - a) 802.11n
 - b) 802.11
 - c) 802.13ac
 - d) 802.32b

- 7) What is the name of an external datacenter that can be used to recover immediately in the event of an outage?
- a) Active DR Site
 - b) Hot DR Site
 - c) Immediate DR Site
 - d) Instant DR
- 8) What is an encryption system called where one key is used to encrypt a message and a separate one is used to decrypt it?
- a) Hybrid encryption
 - b) Symmetric encryption
 - c) Asymmetric encryption
 - d) Bidirectional encryption
- 9) When an exploit successfully gains more permissions in a system, this is called _____.
- a) Privilege Escalation
 - b) Permission Enhancement
 - c) Role Climbing
 - d) Ladder Climbing
- 10) An individual uses a .png file to hide a secret message. The individual has used _____ in the process.
- a) Message Hiding
 - b) Message Spoofing
 - c) Steganography
 - d) Cryptography
- 11) Which of the following devices is used to connect voice, data, pagers, networks, and almost any other conceivable application into a single telecommunications system?
- a) PBX
 - b) Router
 - c) HUB
 - d) Switch
- 12) Any organization that practices sound information security will have a policy that governs the way computers can be used. What is this policy called?
- a) Privacy policy
 - b) Due care policy
 - c) Best practice policy
 - d) Acceptable-use policy

- 13) A software has been installed on a computer that tracks all of the keystrokes entered. What is this software called?
- a) Keytap Tracker
 - b) Key Relay
 - c) Key Logger
 - d) Key Interpreter
- 14) What is the name of the popular operating system based on the Linux kernel?
- a) ME
 - b) Ubuntu
 - c) GNU
 - d) Gnome
- 15) What port is used by SMTP?
- a) 25
 - b) 80
 - c) 22
 - d) 21
- 16) Which attack targets Bluetooth-enabled devices?
- a) Bluespoofing
 - b) Bluejacking
 - c) Tooth pulling
 - d) Bluesnarfing
- 17) Which of the following is a method of cryptography?
- a) Back propagation
 - b) Forward stepping
 - c) Elliptic curve
 - d) Bidirectional aliasing
- 18) Which of the following hijacks a host and demands payment to release the system?
- a) Spyware
 - b) Ransomware
 - c) Bloatware
 - d) Vaporware

- 19) Which of the following RAID setups is *not* fault tolerant?
- a) RAID 1
 - b) RAID 2
 - c) RAID 3
 - d) RAID 0
- 20) This operating system is a security-focused operating system for anonymity and privacy.
- a) Onion
 - b) Tails
 - c) CoreOS
 - d) Plan9
- 21) Which of the following is an example of a PaaS (platform as a service)?
- a) Virtual private network (VPN) host
 - b) A customer relationship management (CRM) suite and full set of integrations
 - c) Serverless code execution platform
 - d) Cloud-based virtual machine provisioning
- 22) Which of the following is considered to be the *weakest* form of security for a wireless network?
- a) WEP
 - b) WAP
 - c) WPA2
 - d) MAC address filtering
- 23) Where is the FBI headquarters located?
- a) New York City
 - b) Austin, TX
 - c) Langley, Virginia
 - d) Washington DC
- 24) What is the name of Microsoft's implementation of the lightweight directory access protocol? (LDAP)
- a) Active Directory
 - b) Azure
 - c) Insight
 - d) Directory Manager

- 25) An access control list (ACL) is composed of _____.
a) Access Control Bodies
b) Security Records
c) Access Control Entries
d) Control Entities
- 26) What does SAM stand for?
a) Secure Acknowledgement Message
b) System Attention Message
c) Security Aware Maintenance
d) Security Account Manager
- 27) _____ is the name of the popular network packet sniffer?
a) WireSniff
b) Wireshark
c) Wiresnarf
d) NetTap
- 28) _____ is the technique used to *remove* the secure layer from HTTPS?
a) SSLStrip
b) SSLDelete
c) Remove SSL
d) xSSL
- 29) What type of system is an intrusion detection system?
a) Passive System
b) Active System
c) Reactive System
d) Latent System
- 30) What type of system is an intrusion prevention system?
a) Passive System
b) Active System
c) Reactive System
d) Latent System
- 31) When an anti-virus identifies a non-malicious file as being harmful software, this is called _____.
a) False Negative
b) False Positive
c) False Assumption
d) Signature Malfunction
- 32) A proxy is a _____.
a) type of physical network device used to control traffic
b) type of software that automates basic tasks
c) host that acts on behalf of another host
d) computer that is able to act as a server

- 33) Which of the following is considered the *most* secure password?
- a) Password123!
 - b) Thisisapassword!@
 - c) Thisisnotapassword
 - d) Th1s1s@P@ssw0rd!
- 34) What vulnerability is present when a webserver does *not* check whether the request came from a trusted client or not?
- a) CSRF
 - b) XSS
 - c) Virus
 - d) IPS
- 35) Which of the following is the term used to define a hacker with malicious intent?
- a) Gray Hat
 - b) Bad Hacker
 - c) Black Hat Hacker
 - d) Cyber Punk
- 36) Which of the following can be used to gather information about a network?
- a) Port Scanning
 - b) Network Mapping
 - c) Packet Sniffing
 - d) All of the above
- 37) What does WAF stand for?
- a) Web Application Firewall
 - b) Windows Authentication File
 - c) Windows Administration Filesystem
 - d) Web Administration Filesystem
- 38) Which of the following allows for configuration of system parameters based on group membership?
- a) Group Justification
 - b) Group-Based Configuration
 - c) Group Policy
 - d) Group Authentication
- 39) Which of the following is the Windows-based webserver included with Windows server operating systems?
- a) Apache
 - b) NGINX
 - c) Tomcat
 - d) IIS

- 40) Which of the following is *most commonly* used to serve Java Server Pages (JSP)?
- a) Tomcat
 - b) IIS
 - c) Apache
 - d) Nginx
- 41) Much of the world's most security-sensitive data (financial, health, personal) is processed by which legacy programming language?
- a) COBOL
 - b) Python
 - c) PHP
 - d) C++
- 42) Which of the following is the name of a popular anti-virus software?
- a) Supersight Data Protector
 - b) Trend Micro
 - c) SuperFixIt Cyberblocker
 - d) Element 7
- 43) Which of the following is the name of the Linux distribution most popular in enterprise environments?
- a) SUSE Linux
 - b) Ubuntu
 - c) Red Hat
 - d) CentOS
- 44) MySQL and Microsoft SQL are examples of what kind of database?
- a) Relational Database Management System (RDBMS)
 - b) Document Store Database
 - c) Columnar or Hypertable Record Database
 - d) Graph Database
- 45) The NSA is responsible for investigating cybercrime within the United States.
- a) True
 - b) False
- 46) Hacking is *never* allowed, even if the target entity gives written consent to perform testing.
- a) True
 - b) False
- 47) Some viruses can change their internal code to avoid signature-based antivirus software.
- a) True
 - b) False

- 48) Computers that are rarely logged onto, but are still connected to the internet, are virtually immune to malicious attacks.
- a) True
 - b) False
- 49) AES is an example of symmetric encryption.
- a) True
 - b) False
- 50) HTTPX is a *more* secure version of HTTPS.
- a) True
 - b) False